

Arium® Ultrapure Water for Trace Analysis

Eliminating background signals in chromatography and mass spectrometry with the Sartorius Arium® Mini Plus

The purity of the solvents used, primarily that of water, is a decisive criterion for interference-free and reproducible analysis by liquid chromatography and for the sensitivity of this method, especially for applications in trace analysis. In a study, different sources of ultrapure water used as eluents were compared in high-performance liquid chromatography with diode-array detectors (HPLC-DAD) and mass spectrometry (MS) systems in various experiments.

Keywords: Arium® Mini Plus, ultrapure water, HPLC-DAD, LC-TOF-MS, liquid chromatography, flavour and fragrance industry

Introduction

In the flavour and fragrance industry, many products are based on natural raw materials, such as vanilla beans, citrus fruits, blossoms and other materials of plant origin. HPLC systems coupled to various detectors (e.g., mass spectrometers, DAD, or refractive index (RI) detectors), are used for quality control of such raw materials in incoming goods inspection and final quality control of outgoing products, as well as in research and development of new products.

A routine analysis performed both in research and in quality control is, for instance, the quantification of vanillin in various samples (e.g., vanilla beans, vanilla extract, vanillin sugar, chocolate, beverages and flavourings) by HPLC-DAD.

Besides the quantification of certain analytes, screening methods for identification of partially unknown

substances contained in raw extracts and natural products, among others, play a significant role in research. Liquid chromatography (LC) systems are predominately used in such methods and are coupled to high-resolution (TOF)-MS instruments.

Avoiding High Background Noise

The purity of the solvents used, primarily that of water, is a decisive criterion for interference-free and reproducible analysis by liquid chromatography and for the sensitivity of this method, especially for applications in trace analysis.

Contaminants in the eluent can result in relatively high baseline noise originating from the detector and thus to a poorer signal-to-noise ratio (S/N) of a peak. In DAD, such contaminants are, for example, organic compounds that absorb light in the UV/VIS range. In LC-MS applications, the concentration of ions (Na^+ , K^+) should be kept as low as possible to prevent the formation of adducts with analytes during ionization. Water contaminants that are enriched in the stationary phase can also be eluted with a higher percentage of organic solvent and occur as potentially co-eluting peaks in the chromatogram.

For these reasons, a number of specially treated and filtered types of ultrapure water are commercially available in different grades (HPLC and LC-MS grades). An alternative to these grades of water that are usually filled in 1- or 2.5 - litre bottles is to use water purified by ultrapure water systems such as the Arium® Mini Plus ultrapure water system (Figure 1). Use of ultrapure water freshly produced by such a system to prepare an eluent for HPLC-DAD and MS systems was compared in various experiments with two commercially available brands of bottled water of LC-MS grade.

For this purpose, the background signal in the chromatogram – usually detectable as a baseline – was examined after a relatively long accumulation phase for each particular water sample used in the chromatographic system and subsequent gradient elution performed on two different detectors (DAD- and

TOF-MS). In addition, representative routine analyses, such as the analysis of vanillin by HPLC-DAD and screening of a natural product by LC-MS, were run with three different sources of ultrapure water as part of the mobile phase and compared.

Production of Ultrapure Water Using Arium® Mini Plus

To produce ultrapure water, Arium® Mini Plus (Figure 1) is directly connected to the tap water feed to purify this water in a two-stage process. In the first stage, this compact system produces pure water, reverse osmosis water (RO water), and in the second stage, ultrapure water. As lower flow rates are reached during RO purification and this stage therefore has a limiting effect on such rates, an Arium® bag is connected as a reservoir between the two stages (flow diagram in Figure 1).

In the first stage of the Arium® Mini Plus system, feed water is passed from the system inlet through a pretreatment cartridge, an RO module, by using a diaphragm pump. The RO module has two outlets, one for the permeate flow and the other for the concentrate flow. The latter flow path is connected to the system's outlet to drain off the water removed from the RO purification stream, "rejected water." The permeate flow is purified RO water (i.e., pure water) that fills the bag and is monitored in the process by a conductivity cell.

In the second downstream stage, the pure water obtained is transported by a further pump out of the bag to the actual purification cartridge for generating ultrapure water.

Here, pure water is transformed into ultrapure water using an optional UV lamp (has an oxidizing at wavelengths of 185 nm and 254 nm, respectively) and by passing through a cartridge filled with active carbon and ion exchange resin. During purification, the quality of ultrapure water is continuously monitored by a second conductivity cell to maintain a conductivity of $0.055 \mu\text{S/cm}$ (corresponds to a resistivity of $18.2 \text{ M}\Omega \times \text{cm}$),

Application Showcase

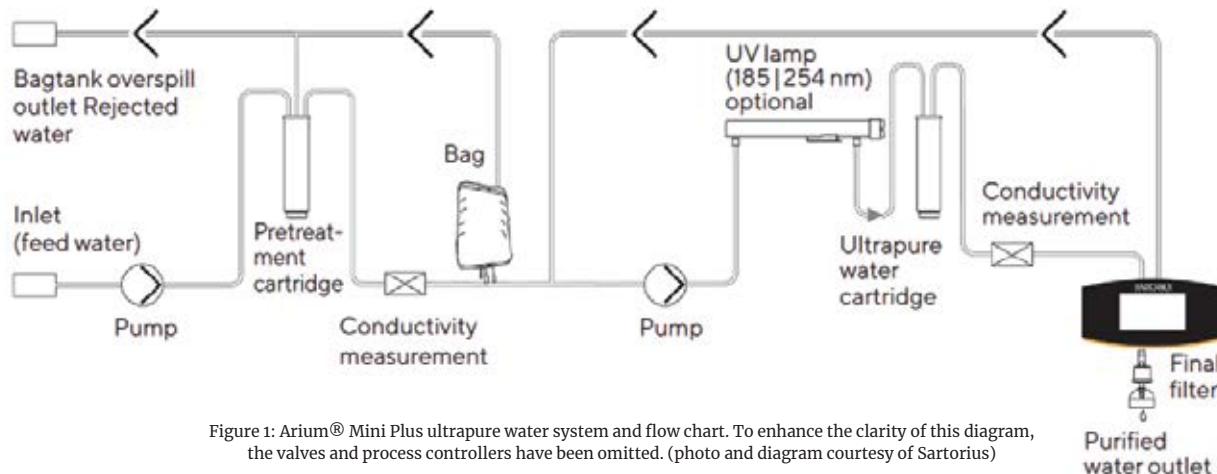


Figure 1: Arium® Mini Plus ultrapure water system and flow chart. To enhance the clarity of this diagram, the valves and process controllers have been omitted. (photo and diagram courtesy of Sartorius)

compensated to 25 °C. Then in the last purification step, purified water is dispensed via a final sterilizing-grade filter.

This process is shown as a schematic diagram in Figure 1.

Materials and Methods

The water sources tested included two commercially available, certified water grades for LC-MS applications (LC-MS grade water A and B) besides ultrapure water freshly produced by the Arium® Mini Plus system. To examine the background signal in the HPLC-DAD and LC-MS chromatograms, the different water sources were each passed through an RP-C18 column, under standard flow conditions and without the addition of modifiers (e.g., formic acid or buffer), for a period of 40 min. for the HPLC-DAD method and 16 min. for the LC-MS method, respectively, to concentrate any contaminants present in each water source (accumulation phase).

Then the potential contaminants were eluted by running a gradient of water used as the solvent to 100% acetonitrile. At the end of each run, the column was reconditioned with the respective water source. This was carried out on two different systems: HPLC-DAD (system 1 supplied by Agilent based in Waldbronn, Germany) and LC-TOF-MS (system 2 supplied by

Waters, Eschborn and Bruker based in Karlsruhe, Germany). The device parameters are listed in Table 1.

In routine analysis, system 1 is used, for example, to quantify vanillin, whereas system 2 is mainly employed for screening of compounds, such as those in natural products. For these two applications, trial runs were performed using the different water sources.

Results

Background Signal in UV and MS Detection

The resulting chromatograms of the trial runs conducted on the HPLC-DAD system (system 1) are shown in Figure 2.

In the top graph (Figure 2A), the chromatograms of the three samples of ultrapure water are overlaid, and the gradient profile is marked.

On the chromatograms, both the commercially available bottled water grades and ultrapure water freshly produced by the Arium® Mini Plus system show similar contaminants that were accumulated in the separation column. At approx. 45 minutes, a broad peak was observed, which exhibits a substantially higher peak area for accumulation using Arium® Mini Plus water and, due to its asymmetry, indicates overlay of several contaminants.

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	System 1	System 2
Device	Agilent 1290 with DAD	Waters Acquity UPLC, Bruker microTOF II
Separation Column	Grom Sapphire C18, 150 x 2.1 mm, 5 µm (Grom analytical + HPLC, Herrenberg, Germany)	Kinetex RP-C18, 100 x 2.1 mm, 1.7 mm (Phenomenex, Aschaffenburg, Germany)
Mobile phase	A: Water B: Acetonitrile	
Gradient	0 min. 100% A 0%B	0 min. 100%A 0%B
	40 min. 100%A 0%B	16 min. 100%A 0%B
	50 min. 0%A 100%B	19 min. 5%A 95%B
	60 min. 0%A 100%B	23 min. 0%A 100%B
	70 min. 100%A 0%B	26 min. 100%A 0%B
		30 min. 100%A 0%B
Temperature	40 °C	50 °C
Flow rate	1 mL/min.	0.55 mL/min.
Detection	200 nm	50-1600 Da (ESI- and ESI+)

Table 1: Device Parameters of Systems 1 and 2 for Analysis of Contaminants Present in the Different Water Sources

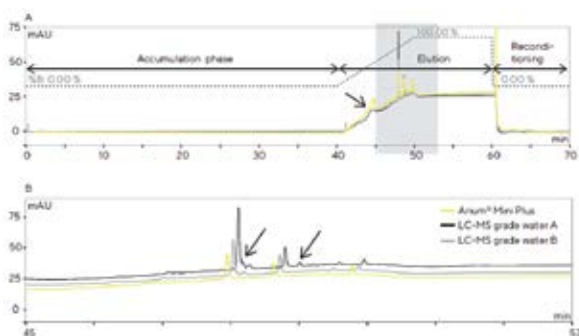


Figure 2: (A) HPLC-DAD chromatograms (detection: 200 nm) after a 40-min. accumulation phase for each of the three water sources tested in the column and subsequent elution of the contaminants performed with acetonitrile. (B) Magnified view of the coloured section in A; differences in the peak profile are identified by arrows.

In the coloured section (45 – 53 min.; magnified view in Figure 2B), there are slight differences in the peak profile.

Here, the profiles for Arrium® Mini Plus water and water B are comparable, whereas in water A, contaminants that cannot be observed in the other chromatograms are detected. This accumulation experiment delivered reproducible results (n = 5).

This trial conducted to examine the background signal after accumulation of the contaminants for the various samples of ultrapure water on an RP-C18 separation column was additionally performed with a

high-resolution TOF-MS system (system 2, Figure 3). After electrospray ionization in positive mode (ESI+), hardly any differences can be seen between the peak profiles (data not shown). By contrast, differences can be seen in the peak profiles obtained in the ESI- mode (Figure 4).

Thus, the chromatogram for Arrium® Mini Plus water in the range of 22 – 25 min. shows fewer peaks of contaminants in comparison to those obtained for the commercially available brands of bottled water.



Figure 3: Waters Acquity UPLC with Bruker microTOF II (location and photo: Symrise)

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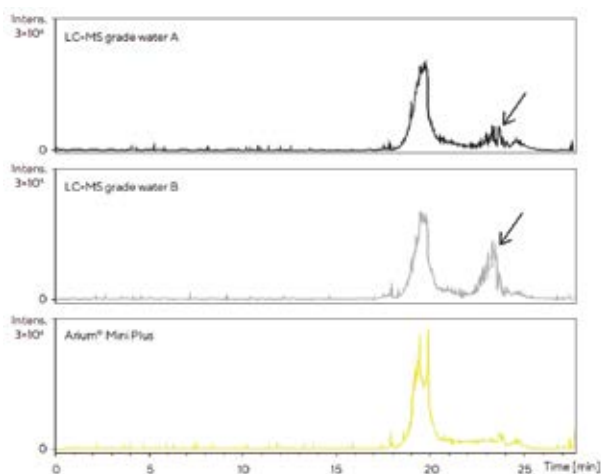


Figure 4: LC-TOF-MS chromatograms after the accumulation phase for each of the three water sources tested in the column and subsequent

In addition, the particular water samples were injected by a syringe pump directly into the TOF-MS system (Bruker microTOF II).

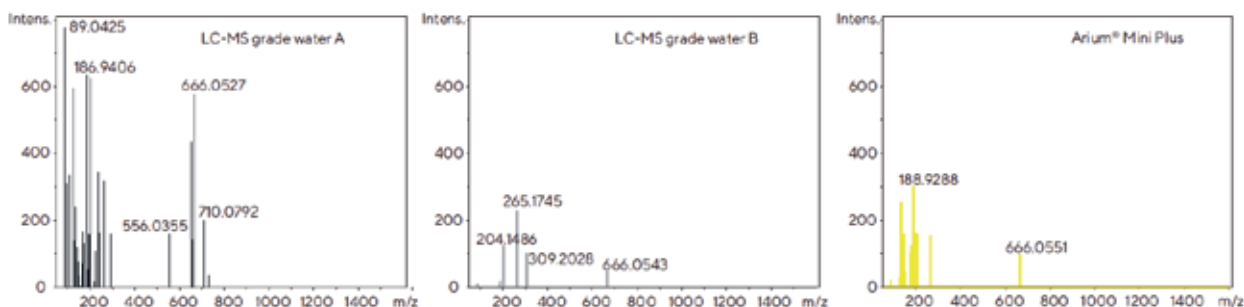


Figure 5: Mass spectra of each of the tested water sources, obtained after direct injection by syringe pump following electrospray ionization in the negative mode (ESI⁻).

Comparison of Water Sources in Routine Analyses

To test the usability of the different ultrapure water sources in routine analyses, these sources were each employed as solvents in sample chromatographic runs.

Quantification of Vanillin in Vanilla Extract

After diluting with methanol (approx. 1:1,000), vanilla extract was analysed using each of the three different water sources as a component of the respective eluents run through system 1.

The vanillin concentration of each injected solution corresponded in this case to approx. 4 µg/mL.

As special experiments are carried out by direct injection just as is generation of reference spectra both in the scan and MS/MS modes, it is also important in these cases that the quantity of interfering ions produced from the solvents used be kept as low as possible. The spectra recorded were averaged by software over a time span of 1 min.

Figure 5 shows examples of the spectra obtained in the ESI mode. By comparison, LC-MS grade water B and water from the Arium® Mini Plus system show fewer signals of potentially interfering ions, whereas LC-MS grade water A generates considerably more signals. This can also be observed in the ESI⁺ mode (data not shown) and supports the observations made in assessing the chromatograms depicted in Figure 4.

The resulting chromatograms (Figure 6A) are nearly congruent, and the peak areas of vanillin do not differ at all.

However, if the vanillin concentration is within the range of the detection limit (9 ng/mL), the background signal, as a result of the water purity, does play a role, and substantial differences can be seen (Figure 6B). In view of the baseline curve and signal-to-noise ratio (S/N) of the vanillin peak, Arium® Mini Plus water and LC-MS grade water B are comparable, whereas the chromatogram for LC-MS grade water A shows a higher baseline and more potentially interfering peaks.

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This confirms the observations made in the experiments described above with regard to the background signals in the HPLC-DAD system and after direct injection into the TOF-MS system.

Screening of Orange Oil Using LC-TOF-MS

Furthermore, the three water sources were tested in a qualitative screening method by high-resolution LC-TOF-MS (system 2) to identify individual compounds in mixtures.

A specific type of orange oil was used as the sample material. The chromatograms are shown in Figure 7. Comparable performance regarding the peak height, peak area, retention time and separation was observed. Likewise, in view of the baseline curve and signal-to-noise ratio, hardly any differences are seen (Figure 7, magnified view).

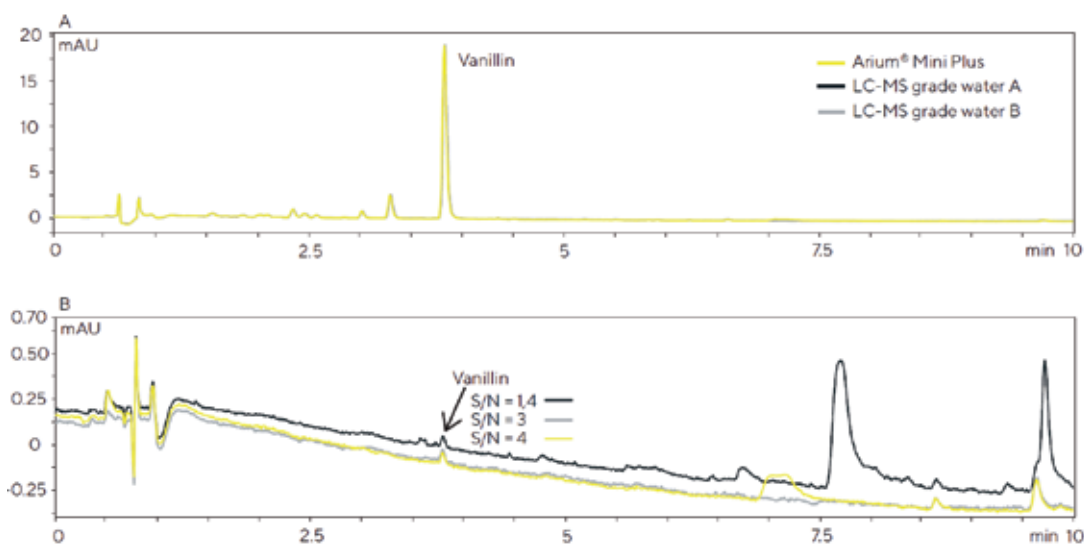


Figure 6: HPLC-DAD chromatograms of a vanilla extract (A) and an aqueous vanillin solution (B, 9 ng/mL) obtained with the different water sources used in the mobile phase (detection: 280 nm; column: Poroshell 120 SB-C18, 2.7 μ m, 100 $\times$ 2.1 mm; eluents: acetonitrile and water with 0.1% formic acid in the gradient mode; flow rate: 0.4 mL/min.).

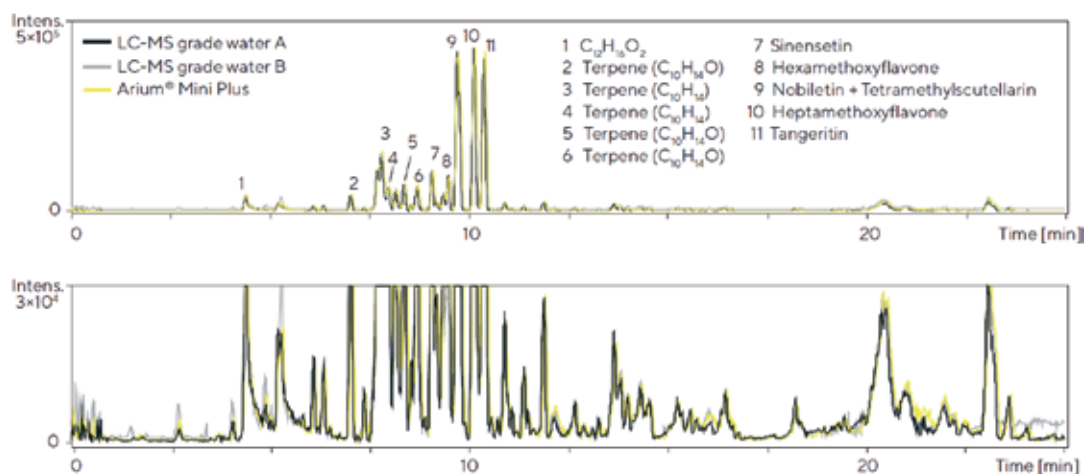


Figure 7: LC-TOF-MS chromatograms (BPC, base peak chromatogram; 70 – 1,600 Da) of an orange oil (diluted 1:20) obtained with the respective water sources in the mobile phase (detection: 50 – 1,600 Da (ESI+); column: Kinetex RP-C18, 1.7 μ m, 100 $\times$ 2.1 mm; eluents: acetonitrile and water, each with 0.1% formic acid in the gradient mode; flow rate: 0.55 mL/min.; lower chromatogram: magnified view).

Discussion

Based on the experiments conducted, it could be shown that Arium® Mini Plus ultrapure water is excellently suited for use in chromatography and mass spectrometry (MS). In view of the potentially co-eluting peaks and the back-Ground signal in UV and MS detection, it was observed that Arium® Mini Plus water is comparable with the tested quality grades of commercially available bottled water.

The background signal, which primarily depends on the purity of the solvent used in chromatographic analysis, must be as low as possible as this signal is highly significant for the sensitivity of the analytical method and for reliable quantification. Besides LC-MS grade water B, Arium® Mini Plus water with a higher S/N excels especially in trace analysis requiring high sensitivity, as shown in the example of the vanillin peak obtained on the HPLC-DAD chromatogram.

Unlike commercially available, bottled ultrapure water, an ultrapure water system offers the considerable advantage of being able to freshly purify water in any quantity on demand. From an economic point of view, this feature is thus a good alternative to purchased ultrapure bottled water. Fresh purification also prevents water from standing in opened bottles for long periods because such water stored in opened bottles can be contaminated by absorption from the laboratory atmosphere^{1,2} and dissolve CO₂ from air, among other contaminants. Organic contaminants in water are detectable by an increase in the TOC level (total organic carbon). At high TOC levels, identification and quantification of trace components can be compromised¹, for instance, by shifts in the baseline^{1,2} or by the occurrence of ghost peaks³.

In addition, if bottled water is stored for relatively long periods, Na-cations, for example, can leach from the glass bottles, which, in turn, can lead to increased formation of adducts during ionization in LC-MS systems. An accordingly lower yield of ions used for

evaluation (usually [M+H]⁺ or [M-H]⁻) ions can have a negative impact on the sensitivity of the method⁴.

The high suitability of fresh ultrapure water, produced by Arium® Pro systems, in different chromatography techniques and the increasing use of these technologies in the most diverse applications will very likely contribute to the growing acceptance and pervasiveness of laboratory water purification systems.

References

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